



Bertil Braun

Applied ML Research Engineer

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🎓 EDUCATION

M.Sc. Informatics | AI specialization

Karlsruhe Institute of Technology (KIT)

2023 – 2024 | Grade 1.2

- Focus: ML, AI, and algorithms
- Thesis: Built an LLM pipeline for competency extraction, fine-tuning, and automated evaluation.
- Completed in one year—half KIT's standard two-year duration.

B.Sc. Informatics

Karlsruhe Institute of Technology (KIT)

2020 – 2023

- Thesis: Studied modular bounded verification for sorting algorithms.

📖 PUBLICATIONS

Engineering Efficient Self-Play Chess: Search, Replay, and Throughput Under Limited Compute

arXiv submission pending

LLM-Light: Artifact-Addressed Execution for a Fixed LLM Experiment Pipeline

arXiv submission pending

(Towards) Scalable Reliable Automated Evaluation with Large Language Models

ACL GEM² Workshop 2025 · 07/2025

Voice-Light: A Full-Duplex Voice Agent with Causal Turn-Taking and Speculative Generation

arXiv:2609.20995 · 09/2026

Identity-Consistent Analysis of Long-Shot Windsurfing Video: An Offline Tracking System

arXiv:2609.24492 · 09/2026

A Graph-Based Control Interface for Traffic Signals on Heterogeneous Road Networks

arXiv:2607.21831 · 07/2026

👤 ABOUT

I like building ML systems where research and engineering are inseparable. I work end to end—from data and training pipelines through evaluation, systems integration, and deployment—across LLM development, agentic applications, reinforcement learning, computer vision, and conversational speech. I completed KIT's M.Sc. Informatics with an AI specialization in one year—half the standard two-year duration—and bring 8+ years of hands-on software development across self-directed engineering, academic research, and industry. I learn new domains quickly without sacrificing the depth required to build rigorous, working systems. I am looking for an ambitious team that will keep pushing me.

📁 EXPERIENCE

Applied AI Consultant

Confidential early-stage client

06/2026 – 07/2026 | London, United Kingdom

Analyzed dataset quality and built automated evaluation, provenance, and annotation workflows, while providing technical input on data-processing and evaluation architecture.

Independent AI Research Engineer

Self-Directed Research & Systems Development

04/2025 – Present | Sindelfingen, Germany

Conducted full-time independent ML research across model development, systems engineering, evaluation, and publication.

- Authored six ML papers and technical reports across LLM, voice, graph-RL, vision, and self-play systems, supported by evaluations and artifacts.
- Built LLM-Light and Voice-Light end to end, covering synthetic data generation, model training, reproducible evaluation, and public deployment.
- Engineered a 22-service agent platform with A2A/MCP orchestration, Postgres event logs, vector/graph memory, and end-to-end observability.
- Scaled AlphaZero chess through distributed self-play and multi-GPU training, reaching 3,251 Stockfish-calibrated Elo under a fixed compute budget.

Karlsruhe Institute of Technology (KIT)

Research Assistant - Computer Science

01/2025 – 04/2025 | Karlsruhe Institute of Technology (KIT)

Conducted research on automated LLM evaluation, resulting in a first-author publication at ACL 2025 Workshop.

CAS Software AG

AI Engineer / Master's Thesis

05/2024 – 12/2024 | Karlsruhe, Germany

Developed a domain-agnostic competency extraction system using LLMs, covering multi-phase pipeline design, fine-tuning, and automated evaluation methods. Work formed the basis of the ACL 2025 publication. Subsequently contributed as AI Engineer to production LLM systems.

TECHNICAL SKILLS

Programming & ML: Python, C++, PyTorch, NumPy

Machine Learning: LLM training and evaluation, reinforcement learning, computer vision, conversational AI

ML Systems: GPU training and inference, distributed pipelines, Docker, FastAPI

Algorithm Engineer

09/2023 – 05/2024 | Karlsruhe, Germany

Worked within a complex in-house constraint satisfaction system (boolean, numerical, and string constraints) for industrial product configurators. Independently researched and prototyped the integration of a Mercedes-specific constraint rule, delivering a viability evaluation and working prototype.

Mercedes-Benz AG

Working Student – Automation Engineering

04/2022 – 04/2023 | Stuttgart, Germany

Built an automation framework used by engineers to set up, run, and evaluate mechanical simulations.

PROJECTS

AlphaZero-Style Chess & Go: Scalable Deep Reinforcement Learning

01/2023 – 10/2026

Distributed self-play, batched tree search, and multi-GPU training in Python/C++.

- Built C++ native chess/Go rules, batched MCTS, CUDA inference, and Stockfish evaluation.
- Generated 250M+ self-play positions across roughly 3M games and 100B+ MCTS simulations.
- Reached 3,251 Stockfish-calibrated Elo under a fixed 2.5-day, \$43 training-compute budget.
- Distilled to a 470k-parameter model—13× smaller—reaching 2,873 Stockfish-calibrated Elo.

LLM-Light – End-to-End LLM Pretraining & Evaluation System

06/2026 – 07/2026

PyTorch-native pipeline for reproducible GPT and MoE experiments.

- Built a config-driven pipeline from data generation through training and evaluation.
- Designed artifact-addressed stages with caching, resume, and durable recovery.
- Integrated GPT/MoE pretraining, DPO post-training, inference, and evaluation.
- Validated on 2.19M synthetic Python samples and multiple training runs.

Voice-Light – Low-Latency Conversational Voice AI

07/2026 – 09/2026

Streaming ASR–LLM–TTS system with natural turn-taking and spoken tool use.

- Built and deployed a full-duplex Nemotron→Qwen→Kyutai voice agent over WebSockets.
- Generated 3,999 tool-use conversations and trained tool-use and turn-taking adapters.
- Improved correct tool decisions from 80.0% to 95.2% with Qwen3-1.7B fine-tuning.
- Measured 758 ms median final-speech-to-server-audio latency across 36 live turns.

Agentic LLM Systems: Durable Runtime & Multi-Agent Orchestration

04/2026 – 06/2026

Resumable tool-using agents, memory, and production-style service orchestration.

- Built a durable coding runtime with resumable workflows and human review gates.
- Integrated agents with tests, git diffs, shell output, and tool execution.
- Deployed a 22-service stack with A2A/MCP, Neo4j, Qdrant, and OpenTelemetry.
- Built Postgres event-log workflows with retries, signals, and child runs.

GNN-Based Traffic Signal Control

03/2026 – 07/2026

Movement-level GNN control interface for heterogeneous road networks.

- Built a SUMO/TraCI environment supporting heterogeneous traffic networks.
- Designed a shared GNN that scores individual traffic movements.
- Mapped scores to variable-size legal phases outside the learned model.
- Evaluated PPO transfer on unseen grids, coverage shifts, and five city graphs.

GybeLock – Multi-Object Tracking & Video Intelligence System

07/2025 – 03/2026

Computer vision system for detecting and tracking windsurfers in raw session footage.

- Built a pipeline for rider detection, tracking, stabilization, and focused playback.
- Trained YOLO detection/pose models and an orientation classifier.
- Reduced excess track fragments from 845 to 42 versus OC-SORT on 41k detections.
- Deployed a React/Firebase frontend, FastAPI backend, and Modal GPU pipeline.